

X-13ARIMA-SEATS Quick Reference for Unix/Linux
Version 1.1
April 22, 2025

Running X-13ARIMA-SEATS on a single series:

`{path1}\x13as {-i} {path2}\filename`

where:

`{-i}` : optional flag that informs X-13ARIMA-SEATS that the named file is an input specification file
`{path2}\ filename.spc` : input specification file
`{path2}\ filename.out` : main output file
`{path2}\ filename.err` : error file
`{path2}\ filename.log` : log file, which gives a summary of user-selected diagnostics
`{path1}\` : path information for the X-13ARIMA-SEATS program (optional)
`{path2}\` : path information for the X-13ARIMA-SEATS input file (optional)

Example:

`/vol/x13as/x13as /vol/sales/retail`

Notes:

- (a) Only the filename (and path, if necessary) is specified, not the extension.
- (b) The program uses this filename to form the filename of other files generated by the program.

Running X-13ARIMA-SEATS on a single series with an alternate output name:

`{path1}\x13as {-i} {path2}\filename1 {-o} {path3}\filename2`

where:

`{-i}` : optional flag that informs X-13ARIMA-SEATS that the named file is an input specification file
`{path2}\ filename1.spc` : input specification file
`{-o}` : optional flag that informs X-13ARIMA-SEATS that the named file is to be used to form the output files
`{path3}\ filename2.out` : main output file
`{path3}\ filename2.err` : error file
`{path3}\ filename2.log` : log file, which gives a summary of user-selected diagnostics
`{path1}\` : path information for the X-13ARIMA-SEATS program (optional)
`{path2}\` : path information for the X-13ARIMA-SEATS input file (optional)
`{path3}\` : path information for the X-13ARIMA-SEATS output file (optional)

Example:

`/vol/x13as/x13as /vol/sales/retail /vol/sales/retail2`

Notes:

- (a) Only the filename (and path, if necessary) is specified, not the extension.
- (b) The program uses the alternate filename to form the filename of other files generated by the program.

Handling spaces in file names for a single series:

`{path1\}x13as {-i} "{path2\}filename"`

where:

`{path2\} filename.spc` : input specification file, enclosed in quotation marks ("
`{path1\}` : path information for the X-13ARIMA-SEATS program (optional)
`{path2\}` : path information for the X-13ARIMA-SEATS input file (optional)

Example:

`/vol/x13as/x13as "/vol/US sales/NE retail"`

Notes:

- (a) The opening and closing quotation marks must fully contain the filenames with no extra spaces, and there must be matching opening and closing quotation marks for each file.
- (b) Quotation marks can be used to handle spaces in the alternate output filename as well.
- (c) Spaces are not allowed before the file extension (`.spc`, `.mta`, etc.)

Running X-13ARIMA-SEATS on more than one series (a spec file for every series):

`{path1\}x13as -m {path2\}metafile`

where:

`-m` : flag that informs X-13ARIMA-SEATS that the named file is a metafile
`{path2\} metafile.mta` : input metafile
`{path2\} metafile.log` : log file, which gives a summary of all the runs for a given metafile
`{path1\}` : path information for the X-13ARIMA-SEATS program (optional)
`{path2\}` : path information for the X-13ARIMA-SEATS metafile (optional)

Example:

`/vol/x13as/x13as -m /vol/sales/allsales`

Notes:

- (a) Only the filename (and path, if necessary) for the metafile is specified, not the extension
- (b) The metafile must have one or two filenames (without extension) per line, separated by a tab or blank spaces. The first filename is the filename of an input specification file. The second (if specified) is the filename used to form the filenames of the output files for the run specified by the corresponding input selection file.
- (c) If only one filename is given on a particular line, the filename of the input specification file is used to generate the names of the output files.
- (d) Up to 10000 input files can be specified in a single metafile.

Running X-13ARIMA-SEATS on more than one series (one spec file run on many series):

`{path1}\x13as {-i} {path2\}filename -d {path3\}metafile`

where:

`{-i}` : optional flag that informs X-13ARIMA-SEATS that the named file is an input specification file
`{path2\} filename.spc` : input specification file
`-d` : the flag that informs X-13ARIMA-SEATS that the named file is a data metafile
`{path3\} metafile.dta` : data metafile
`{path3\} metafile.log` : log file, which gives the summary of all the runs for a given metafile
`{path1\}` : path information for the X-13ARIMA-SEATS program (optional)
`{path2\}` : path information for the X-13ARIMA-SEATS input file (optional)
`{path3\}` : path information for the X-13ARIMA-SEATS data metafile (optional)

Example:

```
/vol/x13as/x13as sales -d /vol/sales/alldata
```

Notes:

- (a) Only the filename (and path, if necessary) for the data metafile is specified, not the extension.
- (b) The metafile must have one or two filenames per line, separated by a tab or blank spaces. The first filename is the name of a data file (including the file extension). The second (if specified) is the filename (without extension) used to form the filenames of the output files for the run specified by the corresponding input specification file.
- (c) If only one filename is given on a particular line, the filename of the data file is used to generate the names of the output files.
- (d) The X-13ARIMA-SEATS options given in the input specification file are applied to the data read in from each of the files given in the data metafile.
- (e) Up to 10000 data files can be specified in a single data metafile.

Handling spaces in file names for a metafile run

`{path1}\x13as -m " {path2\} metafile "`

where:

`-m` : flag that informs X-13ARIMA-SEATS that the named file is a metafile
`{path2\} metafile.mta` : input metafile, enclosed in quotation marks ("
`{path1\}` : path information for the X-13ARIMA-SEATS program (optional)
`{path2\}` : path information for the X-13ARIMA-SEATS metafile (optional)

Examples:

```
/vol/x13as/x13as -m "/vol/US sales/all sales"  
/vol/x13as/x13as "airline model" -d "/vol/US sales/all sales data"
```

Notes:

- (a) Filenames within the metafile or data metafile can also be surrounded by quotes if they have spaces embedded in the name of the file.
- (b) The opening and closing quotation marks must fully contain the filenames with no extra spaces, and there must be matching opening and closing quotation marks for each file.
- (c) Spaces are not allowed before the file extension (`.spc`, `.mta`, etc.)
- (d) All other rules for constructing and running metafiles are the same.

Other options declared at time of execution:

- c : Sum each of the components of a composite adjustment, but only perform modeling or seasonal adjustment on the total.
- g *dirname* : Store graphics metafile and related files for external graphics in the *dirname* directory.
- n : (No tables) Print only the tables specifically requested in the input specification file.
- p : Turn off pagination in the main output file.
- q : Run X-13ARIMA-SEATS in “quiet” mode (warning messages not sent to console).
- r : Use reduced output format for table formats and headers.
- s : Store seasonal adjustment and regARIMA model diagnostics in a file.
- t : Store timing information in the diagnostics file (if -s or -g not specified, will generate diagnostic file)
- v : Check input specification file(s) for errors only; no other processing.
- w : Use wide (132 character) format in main output file.

Examples:

```
/vol/x13as/x13as -i /vol/trade/imports -o /vol/trade/importsRun2 -s
/vol/x13as/x13as /vol/trade/imports /vol/trade/importsRun2 -s
/vol/x13as/x13as -m "/vol/EU trade/all exports" -g "/vol/EU trade/graph" -q -r
```

Notes:

- (a) The first two examples are equivalent; when the input specification file is given as the first argument, the -i flag is assumed. When the alternate output filename is given as the second argument, the -o flag is assumed.
- (b) Options can entered in any order (ie, -n -s is treated the same as -s -n).
- (c) The -v flag should not be used with the -s, -c, -n, -w, -r, -q or -p flags. A warning message will be generated.
- (d) The -c flag can only be used with the -m flag.
- (e) The -m flag cannot be used with the -d flag.
- (f) The -i flag cannot be used with the -m flag.
- (g) The -o flag cannot be used with the -m and -d flags.
- (h) Quotation marks can be used for all filenames that have spaces.

Specs and arguments for the input specification file

Notes:

- (a) Every input specification file must have either a **series** spec or a **composite** (for runs where a composite seasonal adjustment is performed) spec.
- (b) The first spec in any input specification file must be either a **series**, **composite**, or **metadata** spec. If the **metadata** spec is the first spec in the input specification file, then the second spec must be either the **series** or **composite** spec.
- (c) The **series** and **composite** specs cannot be used in the same input file.
- (d) For the arguments given below, when two or more values are connected by the symbol |, only one of the values can be assigned to the argument in a given run.
- (e) Dates are specified as either *year.month* for monthly data or *year.quarter* for quarterly data. For monthly series, the months can be denoted either by integers (1 to 12) or by month abbreviations (**jan**, **feb**, **mar**, **apr**, **may**, **jun**, **jul**, **aug**, **sep**, **oct**, **nov**, **dec**). For quarterly series, only integers (1 to 4) are allowed. A zero can be placed in front of integers from 1 to 9 for padding (for example, 2002.02 is an acceptable date specification for February 2002).
- (f) Anything on a line after a number sign (#) is considered a comment and is ignored by the program.
- (g) Spec names, arguments, keywords, and dates are not case sensitive. For example, **SeasonalMA** and **seasonalma** are treated the same.
- (h) Multiple arguments must be enclosed in parentheses. If an argument accepts multiple values but only one is given, then the parentheses are optional. If an argument accepts only a single value, the value must not be enclosed in parentheses.

- (i) Either double or single quotation marks are acceptable for character arguments.
- (j) The **data** and **file** arguments cannot be used in the same spec.
- (k) The **data** and **format** arguments cannot be used in the same spec.
- (l) Only one of the **automdl**, **pickmdl** and **arima** specs can be used in the same input file.
- (m) The **x11** and **seats** specs cannot be used in the same input file, and the **x11regression** spec cannot be used in conjunction with the **seats** spec.
- (n) Change of regime regression variables can be specified for seasonal (**seasonal**), trigonometric seasonal (**sincos**), trading day (**td**, **tdnolpyear**, **td1coef**, **td1nolpyear**, **tdstock**, or **tdstock1coef**), length-of-month (**lom**), length-of-quarter (**loq**), or leap year (**lpyear**) regression variables. When a change of regime is specified for one of these regression variables, the program will add an additional set of regression variables that is defined as usual before the date of the change of regime, and set to zero for those observations on or after the change of regime date. A change of regime regression variable is specified by appending a valid date surrounded by slashes to the name of a regression variable in the **variables** argument of the **regression** spec. For example, to specify a change of regime in trading day starting June of 1985, put **td/1985.jun/** in the **variables** argument of the **regression** spec.
- (o) **X-13ARIMA-SEATS** will extend the series with one year of forecasts prior to **X-11** seasonal adjustment and three years of forecasts prior to a **SEATS** seasonal adjustment whenever a **regARIMA** model is specified with no **forecast** spec. The only way to specify a seasonal adjustment without forecast extension when a **regARIMA** model is specified is to set **maxlead** = 0 in the **forecast** spec.
- (p) The **function** and **power** arguments cannot be used together in the **transform** spec.
- (q) The **x11regression** spec cannot be used for a series with missing data.
- (r) When **0.per** is entered for the ending date of the **modelspace** argument of the **series** or **composite** specs, the ending date of the model span will be set to be the final occurrence of the period **per** in the span of data analyzed (ie, **modelspace** = (1980.jan,0.dec) will set the ending date of the model span to the last December of the data).
- (s) The **diff** and **maxdiff** arguments in the **automdl** specs cannot be used in the same input file; if they are found in the same file, only the values of **maxdiff** will be used.
- (t) The **period** argument of the **series** and **composite** specs can be any number up through 12, but only certain values (1, 2, 4, 6, and 12 for **SEATS**, 12 and 4 for **X-11**) are allowed when performing a seasonal adjustment of the series.
- (u) Setting **out** = 2 in the **seats** reduces the number of series that can be saved by the **save** argument.
- (v) Arguments which have been designated **Rarely Used Arguments** in the main documentation are given at the end of each spec, with a “#” as the first character of the line.

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INDIVIDUAL SPECS

```

arima{
  ar = (initial coefficients for AR, or fixed values with suffix f, e.g. -.6f)
  ma = (initial coefficients for MA, or fixed values with suffix f, e.g. -.6f)
  model = (p d q)(P D Q)
  title = "  "
}

automdl{
  acceptdefault = yes | no { default : no }
  checkmu = yes | no { default : yes }
  diff = (regular differencing order, seasonal differencing order)
  ljungboxlimit = number { default : 0.95 }
  maxdiff = (maximum regular differencing order, maximum seasonal differencing order) { default: (2,1) }
  maxorder = (maximum nonseasonal ARMA order, maximum seasonal ARMA order) { default: (2,1) }
  mixed = yes | no { default : yes }
  print = See Table 1 for list of table names
  savelog = See Table 2 for list of diagnostics
  seasonaloverdiff = yes | no { default : no }
  # armalimit = limit for t-statistics of ARMA coefficients { default : 1.0 }
  # balanced = yes | no { default : no }
  # exactdiff = yes | no | first { default : first }
  # fcstlim = limit for average forecast error { default: 15.0 }
  # hrinitial = yes | no { default : no }
  # reducecv = amount of reduction for outlier critical value { default : 0.14286 }
  # rejectfcst = yes | no { default : no }
  # urfinal = limit for the final unit root test { default : 1.05 }
}

```

```

check{
  maxlag = number of acf's to print
    { default: 24 for monthly series, 8 for quarterly series }
  print = See Table 1 for list of table names
  qtype = ljungbox | lb | boxpierce | bp { default : ljungbox }
  save = See Table 1 for list of table names
  savelog = See Table 2 for list of diagnostics
#   acflimit = limit for acf/se(acf) in diagnostic output { default: 1.6 }
#   qlimit = limit for Box-Ljung and Box-Pierce Q statistic in diagnostic output { default: 0.05 }
}

composite{
  appendbcst = yes | no { default: no }
  appendfcst = yes | no { default: no }
  decimals = 0 number of output decimals, must be an integer from 0 to 5, inclusive
  modelspan = ( startdate, enddate )
    { default: starting, ending date of the aggregated series }
  name = " "
  print = See Table 1 for list of table names
  save = See Table 1 for list of table names, and
    Table 3 for a list of tables that can be saved as percentages
  savelog = See Table 2 for list of diagnostics
  title = " "
  type = flow | stock { default: series does not have a specific type }
#   indoutlier = yes | no { default: yes }
#   saveprecision = 15 { default: 15 (number of decimals in save tables) }
#   yr2000 = yes | no { default: yes }
}

estimate{
  exact = ma | arma | none { default: arma (exact m.l.e. for all coefficients) }
  maxiter = maximum number of iterations { default: 1500 }
  outofsample = yes | no { default: no }
  print = See Table 1 for list of table names
  save = See Table 1 for list of table names
  savelog = See Table 2 for list of diagnostics
  tol = convergence tolerance { default: 1.0e-5 }
#   file = " "
#   fix = nochange | all | arima | reg | none { default: nochange }
#   parms = fixed | estimated { default: estimated }
}

force{
  lambda = Value of the parameter  $\lambda$  used to determine the weight matrix  $C$  for the regression method
    of forcing the totals of the seasonally adjusted series {  $-3.0 \leq \lambda \leq 3$ , default: 0.0 }
  mode = ratio | diff { default: ratio }
  print = See Table 1 for list of table names
  rho = Value of the AR(1) parameter ( $\rho$ ) used in the regression method {  $0.0 \leq \rho \leq 1.0$ ,
    { default:  $0.9^{12/\text{period}}$  (where period is 12 for monthly series, 4 for quarterly) }
  round = yes | no { default: no }
  save = See Table 1 for list of table names, and
    Table 3 for a list of tables that can be saved as percentages
  start = month or quarter when forcing starts { default: 1st month or quarter }
  target = original | calendaradj | permprioradj | both { default: original }
  type = none | denton | regress { default: none }
  usefcst = yes | no { default: yes }
#   indforce = yes | no { default: yes }
}

```

```

forecast{
  appendbcst = yes | no { default: no }
  appendfcst = yes | no { default: no }
  exclude = number of observations to drop before starting forecasts { default: 0 }
  lognormal = yes | no { default: no }
  maxback = how many backcasts { default: 0 }
  maxlead = how many forecasts { default: one year, or three years if a SEATS adjustment is specified }
  print = See Table 1 for list of table names
  probability = coverage probability of prediction intervals, assuming normality { default: 0.95 }
  save = See Table 1 for list of table names
}

history{
  endtable = ending date of tables for seasonal adjustment revisions histories
  estimates = ( sadj sadjchng trend trendchng seasonal aic fcst arma td)
  fixmdl = yes | no { default: no (model is re-estimated every time) }
  fixreg = ( td holiday user outlier )
  fstep = vector of forecast leads for the out-of-sample forecasts and MSE's { default: (1,period) }
  print = See Table 1 for list of table names
  sadjlags = vector of (up to 5) target lags for revisions history of the seasonally adjusted series
  save = See Table 1 for list of table names
  savelog = See Table 2 for list of diagnostics
  start = starting date of revision history
  target = concurrent | final { default: final }
  trendlags = vector of (up to 5) target lags for revisions history of the trend component
#   fixx11reg = yes | no { default: no }
#   outlier = ( keep | remove auto ) { default: keep }
#   outlierwin = number of observations to test for outliers { default: one year }
#   refresh = yes | no { default: no }
#   transformfcst = yes | no { default: no }
#   x11outlier = yes | no { default: yes }
}

identify{
  diff = ( orders of nonseasonal differencing ) { default: (0) }
  maxlag = number of acf's and pacf's to print { default: 36 for monthly series, 12 for quarterly series }
  print = See Table 1 for list of table names
  save = See Table 1 for list of table names
  sdifff = ( orders of seasonal differencing ) { default: (0) }
}

metadata{
  keys = ( keys for user-defined metadata )
  values = ( corresponding values for user-defined metadata )
}

outlier{
  critical = critical value for outlier testing | (criticalAO, criticalLS, criticalTC) { default: Table 4 }
  lsrun = number of successive level shifts to test for cancellation { must be  $\leq 7$ , default: 0 }
  method = addone | addall { default: addone }
  print = See Table 1 for list of table names
  save = See Table 1 for list of table names
  savelog = See Table 2 for list of diagnostics
  span = ( startdate, enddate )
  types = none | ao | ls | tc | all { default: (ao ls) }
#   almost = number greater than zero used to set threshold for almost outliers { default: 0.5 }
#   tcrate = number between 0 and 1 { default:  $0.70^{(12/\text{period})}$ , no default if period < 4 }
}

```

```

pickmdl {
  bcstlim = limit for average backcast error { default: 20.0 }
  fcstlim = limit for average forecast error { default: 15.0 }
  file = " " { no default – file must be specified }
  identify = all | first { default: first }
  method = first | best { default: first }
  mode = both | fcst { default: fcst }
  outofsample = yes | no { default: no }
  overdiff = limit for overdifferencing { default: 0.9 }
  print = See Table 1 for list of table names
  qlim = limit for probability of Ljung-Box Q { default: 5.0 }
  savelog = See Table 2 for list of diagnostics
}

regression{
  aicdiff = difference needed for AIC-based test to accept regressor { default: 0.0 }
  aictest = ( td | tdnolpyear | td1coef | td1nolpyear | tdstock | tdstock1coef | lpyear | loq | lom
             easter | easterstock | user )
  chi2test = yes | no { default: no }
  chi2testcv = critical value probability for chi2test { must be > 0.0 and < 1.0, default: 0.01 }
  file = " " OR data = ( )
  format = "( valid FORTRAN format )" | "datevalue" | "datevaluecomma" | "free" |
            "freecomma" | "x12save" | "x13save" { default: "free" }
  print = See Table 1 for list of table names
  pvaictest = critical value probability for AIC test { must be > 0.0 and < 1.0,
              { default: use aicdiff to set critical value for AIC testing }
  save = See Table 1 for list of table names
  savelog = See Table 2 for list of diagnostics
  start = date { default: the beginning of the series }
  testalleaster = yes | no { default: no }
  tlimit = Critical value of t-statistic for AO or LS sequence regressors
           { default: AO or LS regressors in sequence not checked for significance }
  user = (names of user-defined regression variable(s))
  usertype = ( constant | seasonal | td | lpyear | lom | loq | user
              ao | ls | so | holiday | holiday2 | holiday3 | holiday4 | holiday5 )
  variables = (const | seasonal | sincos[1 to period/2]
              aodate | lsdate | tcdate | sodate | aosdate-date | lssdate-date
              rpdate-date | tldate-date | qidate-date | qddate-date
              td | tdnolpyear | td1coef | td1nolpyear | tdstock[1 to 31] | tdstock1coef[1 to 31] | lpyear | loq | lom
              easter[0 to 25] | easterstock[1 to 25] | sceaster[1 to 24] | labor[1 to 25] | thank[-8 to 17] )
# b = (fixed values for regression coefficients if specified with suffix f, e.g. -.6f; values specified without a
# suffix f are ignored and these coefficients are estimated by GLS)
# centeruser = mean | seasonal { default: user-defined regressors are not centered }
# eastermeans = yes | no { default: yes }
# noapply = (td | ao | ls | tc | so | holiday | userseasonal | user)
# tcrate = number between 0 and 1 { default: 0.70(12/period), no default if period < 4 }
# trendtc = yes | no { default: no }
}

```

```

seats{
  appendfcst = yes | no { default: no }
  finite = yes | no { default: no }
  hpcycle = yes | no { default: yes }
  hplan = parameter used to compute the modified Hodrick-Prescott filter
  hprmls = yes | no { default: no }
  hptarget = orig | sadj | trend { default: trend }
  noadmiss = yes | no { default: no }
  out = 0 | 1 | 2 { default: 0 }
  print = See Table 1 for list of table names
  printphtrf = 0 | 1 { default: 0 }
  qmax = limit for the Ljung-Box Q statistic { default: 50 }
  save = See Table 1 for list of table names; Table 3 for tables that can be saved as percentages;
        Table 7 for tables that can only be saved; and Table 5 for output from the original SEATS program.
  savelog = See Table 2 for list of diagnostics
  statseas = yes | no { default: yes }
  tabtables = Character string with components to save – see Table 6 for list of components
#   bias = -1 | 0 | 1 { default: 1 }
#   epsiv = Convergence criteria for ARIMA estimation within the SEATS module { default: 0.001 }
#   epsphi = allocated  $\phi(\mathbf{B})$  to the seasonal if its frequency differs from the seasonal frequencies
#           by less than epsphi degrees { default: 2 }
#   imean = yes | no { default: no }
#   maxit = Number of iterations allowed for ARIMA estimation within SEATS module { default: 20 }
#   rmod = limit for the modulus of an AR root. { default: 0.80 }
#   xl = limit for AR modulus test. { default: 0.99 }
}

series{
  appendbcst = yes | no { default: no }
  appendfcst = yes | no { default: no }
  comptype = none | add | sub | mult | div { default: none }
  compwt = any number >0 { default: 1 }
  decimals = 0 { number of output decimals, must be an integer from 0 to 5, inclusive }
  file = " " OR data = ( )
  format = "( valid FORTRAN format )" | "1r" | "2r" | "1l" | "2l" | "2l2" | "cs" | "cs2" | "free" |
           "freecomma" | "datevalue" | "datevaluecomma" | "tramo" | "x12save" | "x13save" { default: "free" }
  modelspan = ( startdate, enddate ) { default: starting, ending date of span }
  name = " "
  period = 12 | 4 | (SEATS adjustments only: 6 | 2 | 1) { default: 12 }
  precision = 0 { number of input decimals, must be an integer from 0 to 5, inclusive }
  print = See Table 1 for list of table names
  save = See Table 1 for list of table names
  span = ( startdate, enddate )
  start = date
  title = " " { default: "X-13A-S run for name" }
  type = flow | stock { default: series does not have a specific type }
#   divpower = 4 { rescale series by power of 10, must be integer from -9 to 9 }
#   missingcode = any number { default: -99999. }
#   missingval = any number { default: 1000000000. }
#   saveprecision = 10 { default: 15 (number of decimals in save tables) }
#   trimzero = yes | no | span { default: yes }
#   yr2000 = yes | no { default: yes }
}

```

```

slidingspans{
  cutchn = any number >0 { default: 3.0 }
  cutseas = any number >0 { default: 3.0 }
  cuttd = any number >0 { default: 2.0 }
  fixmdl = yes | no | clear { default: yes (model is fixed every span) }
  fixreg = ( td holiday user outlier )
  length = length of sliding span { default: selected by program }
  numspans = number of sliding spans (between 2 and 4, inclusive) { default: selected by program }
  outlier = yes | keep | remove { default: keep }
  print = See Table 1 for list of table names
  save = See Table 1 for list of table names
  savelog = See Table 2 for list of diagnostics
  start = starting date of sliding spans comparisons { default: selected by program }
#  additivesa = percent | difference { default: difference }
#  fixx11reg = yes | no { default: yes }
#  x11outlier = yes | no { default: yes }
}

spectrum{
  logqs = yes | no { default: no }
  print = See Table 1 for list of table names; Table 8 for tables that can only be saved.
  qcheck = yes | no { default: no }
  save = See Table 1 for list of table names
  savelog = See Table 2 for list of diagnostics
  start = date { default: eight years before end of span }
  tukey120 = yes | no { default: yes }
#  decibel = yes | no { default: yes }
#  difference = yes | no | first { default: yes,  $\max(d + D - 1, 1)$  order of differencing }
#  maxar = 30 { set the maximum order of the AR spectrum (from 1 to 30), default: 30 }
#  peakwidth = 3 { determines width of band used to determine spectral peaks, default: 1 }
#  robustsa = yes | no { default: yes }
#  series = original | a1 | outlieradjoriginal | a19 | adjoriginal | b1 | modoriginal | e1 { default: b1 }
#  siglevel = 6 { determines level (number of stars) used to determine visual peaks, default: 6 }
#  type = arspec | periodogram { default: arspec }
}

transform{
  adjust = lom | log | lpyear
  aicdiff = AICC difference needed to accept no transformation
    { default: -2.0 for monthly and quarterly series, 0.0 otherwise }
  file = " " OR data = ( )
  format = "( valid FORTRAN format )" | "1r" | "2r" | "1l" | "2l" | "2l2" | "cs" | "cs2" | "free" |
    "freecomma" | "datevalue" | "datevaluecomma" | "tramo" | "x12save" | "x13save" { default: "free" }
  function = none | log | sqrt | inverse | logistic | auto { default: none }
  mode = percent | ratio | diff { default: percent }
  name = " "
  power = power for Box-Cox power transformation { default: 1 (no transformation) }
  precision = 0 { number of input decimals, must be an integer from 0 to 5, inclusive }
  print = See Table 1 for list of table names
  save = See Table 1 for list of table names
  savelog = See Table 2 for list of diagnostics
  start = date { default: beginning of the series }
  title = " "
  type = ( temporary permanent ) { default: permanent }
#  constant = positive number to be added to the original series { default: no constant }
#  trimzero = yes | no | span { default: yes }
}

```

```

x11 {
  appendbcst = yes | no { default: no }
  appendfcst = yes | no { default: no }
  final = ao | ls | tc | user { default: all listed effects kept in final seasonally adjusted series }
  mode = mult | add | logadd | pseudoadd { default: mult }
  print = See Table 1 for list of table names
  save = See Table 1 for list of table names, and Table 3 for a list of tables that can be saved as percentages
  savelog = See Table 2 for list of diagnostics
  seasonalma = x11default | s3x1 | s3x3 | s3x5 | s3x9 | s3x15 | stable | msr { default: msr }
  signalim = (1.5 2.5) | (lower and upper sigma limits, both > 0)
  title = " "
  trendma = any odd number less than or equal to 101 { default: automatic trend selection }
  type = sa | summary | trend { default: sa }
#  calendarsigma = all | signif | select | none { default: none }
#  centerseasonal = yes | no { default: no }
#  excludefcst = yes | no { default: no }
#  keepholiday = yes | no { default: no }
#  print1stpass = yes | no { default: no }
#  sfshort = yes | no { default: no }
#  sigmavec = list of months to be grouped together when calendarsigma = select
#  trendic = any number > 0 { default : depends on what is entered for trendma }
#  true7term = yes | no { default: no }
}

x11regression {
  aicdiff = difference needed for AIC-based test to accept regressor { default: 0.0 }
  aictest = (td | td1coef | tdstock | tdstock1coef easter user)
  critical = critical value for AO outlier testing { default: depends on length of span, see Table 4 }
  file = " " OR data = ( )
  format = "( valid FORTRAN format )" | "datevalue" | "datevaluecomma" | "freecomma"
    | "free" | "tramo" | "x12save" | "x13save" { default: "free" }
  outliermethod = addone | addall { default: addone }
  outlierspan = ( startdate, enddate )
  print = See Table 1 for list of table names
  prior = yes | no { default: no }
  save = See Table 1 for list of table names
  savelog = See Table 2 for list of diagnostics
  sigma = any number > 0 { default: 2.5 }
  span = ( startdate, enddate ) { default: starting, ending date of span }
  start = date { default: the beginning of the series }
  tdprior = (td weight for each day of week) { default: no prior trading day }
  user = (names of user-defined regression variable(s))
  usertype = ( td holiday user )
  variables = (td | td1coef | tdstock[1 to 31] | tdstock1coef[1 to 31] aodate
    easter[0 to 25] | sceaster[1 to 24] labor[1 to 25] thank[-8 to 17] )
#  almost = number greater than zero used to set threshold for almost outliers { default: 0.5 }
#  b = (fixed values for regression coefficients if specified with suffix f, e.g. -.6f; values specified without a
#  suffix f are ignored and these coefficients are estimated by OLS)
#  centeruser = mean | seasonal { default: user-defined regressors are not centered }
#  eastermeans = yes | no { default: yes }
#  forcecal = yes | no { default: no }
#  noapply = (td holiday)
#  reweight = yes | no { default: no }
#  umfile = " " { file containing user-defined mean } OR umdata = ( ) { user-defined mean }
#  umformat = "( valid FORTRAN format )" | "1r" | "2r" | "11" | "21" | "2l2" | "cs" | "cs2" |
#  "datevalue" | "datevaluecomma" | "free" | "freecomma" | "tramo" | "x12save" | "x13save"
#  { default: "free" }
#  umname = " " { name of the user-defined mean }
#  umprecision = 0 { number of input decimals, must be an integer from 0 to 5, inclusive }

```

```
# umstart = date { default: the beginning of the series }
# umtrimzero = yes | no | span { default: yes }
}
```

Table 1: **Tables Printed or Saved by X-13ARIMA-SEATS**

Name	Abbrev.	Save Table?	Brief	Default	Spec
autochoice	ach		+	+	automdl
autochoicemdl	amd				automdl
autodefaulttests	adt				automdl
autofinaltests	aft				automdl
autoljungboxtest	alb				automdl
bestfivemdl	b5m				automdl
header	hdr		+	+	automdl
unitroottest	urt		+	+	automdl
unitroottestmdl	urm				automdl
acf	acf	+		+	check
acfplot	acp			+	check
acfsquared	ac2	+		+	check
acfsquaredplot	ap2			+	check
durbinwatson	dw			+	check
friedmantest	frt			+	check
histogram	hst			+	check
normalitytest	nrm			+	check
pacf	pcf	+			check
pacfplot	pcp				check
adjcompositeplot	b1p				composite
adjcompositesrs	b1	+	+	+	composite
calendaradjcomposite	cac	+			composite
compositeplot	cmp				composite
compositesrs	cms	+	+	+	composite
header	hdr		+	+	composite
indadjstot	iaa	+	+	+	composite
indadjstfac	iaf	+		+	composite
indadjstmentratio	il8	+			composite
indaoutlier	iao	+		+	composite
indcalendar	ica	+		+	composite
indcalendaradjchanges	ie8	+		+	composite
indforcefactor	iff	+	+	+	composite
indftstd8	idf				composite
indirregular	iir	+		+	composite
indirregularplot	iip				composite
indlevelshift	ils	+		+	composite
indmcdmovavg	if1	+			composite
indmodirr	ie3	+			composite
indmodoriginal	ie1	+			composite
indmodsadj	ie2	+			composite
indqstat	if3				composite
indreplacsi	id9	+		+	composite
indresidualseasf	irf			+	composite
indrevsachanges	ifa	+		+	composite
indrndsachanges	ifr	+		+	composite
indrobustsa	iee	+			composite
indsachanges	ie6	+		+	composite
indsadjround	irn	+	+	+	composite
indseasadj	isa	+	+	+	composite
indseasadjplot	iap				composite
indseasonal	isf	+	+	+	composite

Table 1: X-13ARIMA-SEATS Tables (continued)

Name	Abbrev.	Save Table?	Brief	Default	Spec
indseasonaldiff	isd	+	+	+	composite
indseasonalplot	isp				composite
indtest	itt		+	+	composite
indtotaladjustment	ita	+			composite
indtrend	itn	+		+	composite
indtrendchanges	ie7	+		+	composite
indtrendplot	itp				composite
indunmodsi	id8	+		+	composite
indx11diag	if2				composite
indyrtotals	ie4				composite
origchanges	ie5	+		+	composite
origwindsaplot	ie0				composite
outlieradjcomposite	oac	+			composite
prioradjcomposite	ia3	+			composite
ratioplotindsa	ir2				composite
ratioplotorig	ir1				composite
armacmatrix	acm	+			estimate
averagefcsterr	afc			+	estimate
estimates	est	+	+	+	estimate
iterationerrors	ite				estimate
iterations	itr	+			estimate
lformulas	lkf				estimate
lkstats	lks	+	+	+	estimate
model	mdl	+	+	+	estimate
options	opt			+	estimate
regcmatrix	rcm	+			estimate
regressioneffects	ref	+			estimate
regressionresiduals	rrs	+			estimate
residuals	rsd	+			estimate
roots	rts	+			estimate
forcefactor	ffc	+	+	+	force
revsachanges	e6a	+		+	force
rndsachanges	e6r	+		+	force
saround	rnd	+	+	+	force
seasadjtot	saa	+	+	+	force
backcasts	bct	+			forecast
forecasts	fct	+		+	forecast
transformed	ftt	+		+	forecast
transformedbcst	btr	+			forecast
variances	fvr	+			forecast
armahistory	amh	+	+	+	history
chngestimates	che	+			history
chngrevisions	chr	+		+	history
chngsummary	chs		+	+	history
fcsterrors	fce	+	+	+	history
fcsthistory	fch	+			history
header	hdr		+	+	history
indsaestimates	iae	+			history
indsarevisions	iar	+		+	history
indsasummary	ias		+	+	history
lkhdhistory	lkh	+	+	+	history
outlierhistory	rot	+	+	+	history
saestimates	sae	+			history
sarevisions	sar	+		+	history
sasummary	sas		+	+	history
seatsmdlhistory	smh	+	+	+	history

Table 1: X-13ARIMA-SEATS Tables (continued)

Name	Abbrev.	Save Table?	Brief	Default	Spec
sfestimates	sfe	+			history
sfilterhistory	sfh	+			history
sfrevisions	sfr	+		+	history
sfsummary	sfs		+	+	history
tdhistory	tdh	+	+	+	history
trendchngeestimates	tce	+			history
trendchngrevisions	tcr	+		+	history
trendchnsummary	tcs		+	+	history
trendestimates	tre	+			history
trendrevisions	trr	+		+	history
trendsummary	trs		+	+	history
acf	iac	+	+	+	identify
acfplot	acp			+	identify
pacf	ipc	+	+	+	identify
pacfplot	pcp			+	identify
regcoefficients	rgc				identify
finaltests	fts	+			outlier
header	hdr			+	outlier
iterations	oit	+			outlier
temporaryls	tls		+	+	outlier
tests	ots				outlier
header	hdr		+	+	pickmdl
pickmdlchoice	pch		+	+	pickmdl
usermodels	umd		+	+	pickmdl
aictest	ats		+	+	regression
aoutlier	ao	+	+	+	regression
chi2test	cts		+	+	regression
dailyweights	tdw				regression
holiday	hol	+	+	+	regression
levelshift	ls	+	+	+	regression
outlier	otl	+	+	+	regression
regressionmatrix	rmx	+			regression
regseasonal	a10	+	+	+	regression
seasonaloutlier	so	+	+	+	regression
temporarychange	tc	+	+	+	regression
tradingday	td	+	+	+	regression
transitory	a13	+	+	+	regression
userdef	usr	+	+	+	regression
adjustfac	s16	+			seats
adjustmentratio	s18	+			seats
cycle	cyc	+			seats
difforiginal	dor	+			seats
diffseasonaladj	dsa	+			seats
difftrend	dtr	+			seats
irregular	s13	+			seats
irregularoutlieradj	se3	+			seats
longtermtrend	ltt	+			seats
seasadjconst	sec	+			seats
seasonal	s10	+			seats
seasonaladj	s11	+			seats
seasonaladjfcstdecomp	afd	+			seats
seasonaladjoutlieradj	se2	+			seats
seasonalfcstdecomp	sfd	+			seats
seasonalsum	ssm	+			seats
seriesfcstdecomp	ofd	+			seats
totaladjustment	sta	+			seats

Table 1: X-13ARIMA-SEATS Tables (continued)

Name	Abbrev.	Save Table?	Brief	Default	Spec
transitory	s14	+			seats
transitoryfcstdecomp	yfd	+			seats
trend	s12	+			seats
trendadjls	stl	+			seats
trendconst	stc	+			seats
trendfcstdecomp	tfd	+			seats
adjoriginal	b1	+	+	+	series
adjorigplot	b1p				series
calendaradjorig	a18	+			series
header	hdr		+	+	series
outlieradjorig	a19	+			series
savefile	sav		+	+	series
seriesmvdj	mv	+	+	+	series
seriesplot	alp				series
span	a1	+	+	+	series
specfile	spc	+	+	+	series
chngs spans	chs	+			slidingspans
factorm means	fmn			+	slidingspans
header	hdr		+	+	slidingspans
indchngs spans	cis	+			slidingspans
indfactorm means	fmi		+	+	slidingspans
indpercent	pci		+	+	slidingspans
inds spans	ais	+			slidingspans
indsf spans	sis	+			slidingspans
indsummary	smi			+	slidingspans
indychngs spans	yis	+			slidingspans
indypercent	piy				slidingspans
indyysummary	siy				slidingspans
percent	pct		+	+	slidingspans
s spans	ads	+			slidingspans
sf spans	sfs	+			slidingspans
ssftest	ssf				slidingspans
summary	sum			+	slidingspans
td spans	tds	+			slidingspans
ychngs spans	ycs	+			slidingspans
yypercent	pcy				slidingspans
yysummary	suy				slidingspans
npsa	npa				spectrum
npsa ind	npi				spectrum
qcheck	qch		+	+	spectrum
qs	qs		+	+	spectrum
q sind	qsi		+	+	spectrum
speccomposite	is0	+	+	+	spectrum
specextresiduals	ser	+	+	+	spectrum
specindirr	is2	+	+	+	spectrum
specindsa	is1	+	+	+	spectrum
specirr	sp2	+	+	+	spectrum
specorig	sp0	+	+	+	spectrum
specresidual	spr	+		+	spectrum
specsa	sp1	+	+	+	spectrum
specseatsirr	s2s	+	+	+	spectrum
specseatssa	s1s	+	+	+	spectrum
tukeypeaks	tpk		+	+	spectrum
aictransform	tac		+	+	transform
perm prior	a2p	+			transform
perm prior adjusted	a3p	+			transform

Table 1: X-13ARIMA-SEATS Tables (continued)

Name	Abbrev.	Save Table?	Brief	Default	Spec
permprioradjustedptd	a4p	+			transform
prior	a2	+	+	+	transform
prioradjusted	a3	+			transform
prioradjustedptd	a4d	+			transform
seriesconstant	alc	+	+	+	transform
seriesconstantplot	acp				transform
tempprior	a2t	+			transform
transformed	trn	+			transform
adjoriginalc	c1	+			x11
adjoriginald	d1	+			x11
adjustdiff	fad	+	+	+	x11
adjustfac	d16	+	+	+	x11
adjustmentratio	e18	+		+	x11
autosf	asf				x11
biasfactor	bcf	+			x11
calendar	d18	+	+	+	x11
calendaradjchanges	e8	+		+	x11
combholiday	chl	+	+	+	x11
extreme	c20	+			x11
extremeb	b20	+			x11
ftestb1	b1f				x11
ftestd8	d8f				x11
irregular	d13	+		+	x11
irregularadjao	ira	+			x11
irregularb	b13	+			x11
irregularc	c13	+			x11
irregularplot	irp				x11
irrwt	c17	+		+	x11
irrwtb	b17	+			x11
mcdmovavg	f1	+			x11
modirregular	e3	+			x11
modoriginal	e1	+			x11
modseasadj	e2	+			x11
modsic4	c4	+			x11
modsid4	d4	+			x11
movseasrat	d9a			+	x11
origchanges	e5	+		+	x11
origwsaplot	e0				x11
qstat	f3				x11
ratioptorig	ra1				x11
ratioplotsa	ra2				x11
replacsi	d9	+		+	x11
replacsib4	b4				x11
replacsib9	b9				x11
replacsic9	c9	+			x11
residualseasf	rsf				x11
robustsa	e11	+			x11
sachanges	e6	+		+	x11
seasadj	d11	+	+	+	x11
seasadjb11	b11	+			x11
seasadjb6	b6	+			x11
seasadjc11	c11	+			x11
seasadjc6	c6	+			x11
seasadjconst	sac	+	+	+	x11
seasadjd6	d6	+			x11
seasadjplot	sap				x11

Table 1: X-13ARIMA-SEATS Tables (continued)

Name	Abbrev.	Save Table?	Brief	Default	Spec
seasonal	d10	+	+	+	x11
seasonaladjregsea	ars	+	+	+	x11
seasonalb10	b10	+			x11
seasonalb5	b5	+			x11
seasonalc10	c10	+			x11
seasonalc5	c5	+			x11
seasonald5	d5	+			x11
seasonaldiff	fsd	+	+	+	x11
seasonalplot	sfp				x11
sib3	b3	+			x11
sib8	b8	+			x11
tdadjorig	c19	+			x11
tdadjorigb	b19	+			x11
tdaytype	tdy		+	+	x11
totaladjustment	tad	+		+	x11
trend	d12	+		+	x11
trendadjls	tal	+			x11
trendb2	b2	+			x11
trendb7	b7	+			x11
trendc2	c2	+			x11
trendc7	c7	+			x11
trendchanges	e7	+		+	x11
trendconst	tac	+			x11
trendd2	d2	+			x11
trendd7	d7	+			x11
trendplot	trp				x11
unmodsi	d8	+		+	x11
unmodsiox	d8b	+		+	x11
x11diag	f2				x11
yrtotals	e4	+		+	x11
calendar	xca	+	+	+	x11regression
calendarb	bxc	+			x11regression
combcalendar	xcc	+	+	+	x11regression
combcalendarb	bcc	+			x11regression
combtradingday	c18	+	+	+	x11regression
combtradingdayb	b18	+			x11regression
extremeval	c14	+		+	x11regression
extremevalb	b14	+			x11regression
holiday	xhl	+	+	+	x11regression
holidayb	bxb	+			x11regression
outlierfinaltests	xft				x11regression
outlierhdr	xoh			+	x11regression
outlieriter	xoi	+			x11regression
outliertests	xot				x11regression
priortd	a4	+	+	+	x11regression
tradingday	c16	+	+	+	x11regression
tradingdayb	b16	+			x11regression
x11reg	c15	+		+	x11regression
x11regb	b15	+			x11regression
xaictest	xat		+	+	x11regression
xregressioncmatrix	xrc	+			x11regression
xregressionmatrix	xrm	+			x11regression

Table 2: Diagnostics Saved to Log File by X-13ARIMA-SEATS

Name	Abbrev.	Spec	Name	Abbrev.	Spec
alldiagnostics	all	automdl	chi2test	cts	regression
automodel	amd	automdl	seatsmodel	smd	seats
autodiff	adf	automdl	x13model	xmd	seats
bestfivemdl	b5m	automdl	durbinwatson	dws	seats
mean	mu	automdl	friedman	frs	seats
alldiagnostics	all	check	normalitytest	nrm	seats
boxpierceq	bpq	check	totalsquarederror	tse	seats
durbinwatson	dw	check	componentvariance	cvr	seats
friedmantest	frt	check	concurrentesterror	cee	seats
ljungboxq	lbq	check	percentreductionse	prs	seats
normalitytest	nrm	check	averageabsdiffannual	aad	seats
seasftest	sft	check	seasonalsignif	ssg	seats
tdftest	tft	check	overunderestimation	oue	seats
indfstabled8	id8	composite	alldiagnostics	all	seats
indicratio	iir	composite	percent	pct	slidingspans
indidseasonal	iid	composite	percents	pcs	slidingspans
indm1	im1	composite	dirnpsa	dnp	spectrum
indm10	imt	composite	dirpeaks	dpk	spectrum
indm11	ime	composite	dirqs	dqs	spectrum
indm2	im2	composite	dirtykeypeaks	dtp	spectrum
indm3	im3	composite	indnpsa	inp	spectrum
indm4	im4	composite	indpeaks	ipk	spectrum
indm5	im5	composite	indqs	iqs	spectrum
indm6	im6	composite	indtukeypeaks	itp	spectrum
indm7	im7	composite	npsa	npa	spectrum
indm8	im8	composite	peaks	spk	spectrum
indm9	im9	composite	qcheck	qch	spectrum
indmovingseasf	isf	composite	qs	qs	spectrum
indmovingseasratio	isr	composite	tukeypeaks	tpk	spectrum
indq	iq	composite	alldiagnostics	all	spectrum
indq2	iq2	composite	autotransform	atr	transform
indtest	itt	composite	fstableb1	fb1	x11
alldiagnostics	all	composite	fstabled8	fd8	x11
aic	aic	estimate	icratio	icr	x11
aicc	acc	estimate	idseasonal	ids	x11
averagefcsterr	afc	estimate	m1	m1	x11
bic	bic	estimate	m10	m10	x11
hannanquinn	hq	estimate	m11	m11	x11
roots	rts	estimate	m2	m2	x11
alldiagnostics	all	estimate	m3	m3	x11
aveabsrevchnge	ach	history	m4	m4	x11
aveabsrevindsa	iaa	history	m5	m5	x11
aveabsrevsa	asa	history	m6	m6	x11
aveabsrevsf	asf	history	m7	m7	x11
aveabsrevsfproj	asp	history	m8	m8	x11
aveabsrevtrend	atr	history	m9	m9	x11
aveabsrevtrendchnge	atc	history	movingseasf	msf	x11
avesumsqfcsterr	afe	history	movingseasratio	msr	x11
alldiagnostics	all	history	q	q	x11
identified	id	outlier	q2	q2	x11
automodel	amd	pickmdl	alldiagnostics	all	x11
aictest	ats	regression	aictest	ats	x11regression

<i>name</i>	<i>short</i>	<i>spec</i>	<i>description of table</i>
indadjustfacpct	ipf	composite	indirect combined adjustment factors expressed as percentages if appropriate
indcalendaradjchangespct	ip8	composite	percent changes in original series adjusted for calendar effects
indirregularpct	ipi	composite	indirect irregular component expressed as percentages if appropriate
indrevsachangespct	ipa	composite	percent changes for indirect seasonally adjusted series with forced yearly totals
indrndsachangespct	ipr	composite	percent changes for rounded indirect seasonally adjusted series
indsachangespct	ip6	composite	percent changes for indirect seasonally adjusted series
indseasonalpct	ips	composite	indirect seasonal component expressed as percentages if appropriate
indtrendchangespct	ip7	composite	percent changes for indirect trend component
origchangespct	ip5	composite	percent changes for composite series
revsachangespct	p6a	force	percent changes in seasonally adjusted series with forced yearly totals
rndsachangespct	p6r	force	percent changes in rounded seasonally adjusted series
adjustfacpct	psa	seats	combined adjustment factors, expressed as percentages if appropriate
irregularpct	psi	seats	final irregular component, expressed as percentages if appropriate
seasonalpct	pss	seats	final seasonal factors, expressed as percentages if appropriate
transitorypct	psc	seats	final transitory component, expressed as percentages if appropriate
adjustfacpct	paf	x11	combined adjustment factors, expressed as percentages if appropriate
calendaradjchangespct	pe8	x11	percent changes in original series adjusted for calendar factors
irregularpct	pir	x11	final irregular component, expressed as percentages if appropriate
origchangespct	pe5	x11	percent changes in the original series
sachangespct	pe6	x11	percent changes in seasonally adjusted series
seasonalpct	psf	x11	final seasonal factors, expressed as percentages if appropriate
trendchangespct	pe7	x11	percent changes in final trend cycle

Name gives the name of each plot for use with the **save** argument.

Short gives a short name for these tables.

Spec indicates the corresponding spec for each table.

Table 3: **Tables Savable as Percentages with save Argument**

Number of Observations Tested	Outlier Critical Value	Number of Observations Tested	Outlier Critical Value
1	1.9600	48	3.6273
2	2.2365	72	3.7323
3	2.4449	96	3.8007
4	2.6180	120	3.8508
5	2.7455	144	3.8898
6	2.8433	168	3.9169
7	2.9215	192	3.9217
8	2.9859	216	3.9484
9	3.0403	240	3.9714
10	3.0871	264	4.0093
11	3.1280	288	4.0253
12	3.1643	312	4.0398
24	3.4194	336	4.0529
36	3.5458	360	4.0650

Table 4: **Default Critical Values for Outlier Identification**

<i>SEATS file</i>	<i>X-13A-S extension</i>	<i>Contents of file</i>
<code>rogtable.out</code>	<code>.rog</code>	Selected statistics from the growth rate output
<code>summarys.txt</code>	<code>.sum</code>	Summary information and diagnostics from SEATS adjustment
<code>table-s.out</code>	<code>.tbs</code>	Annotated listing of the series, the seasonally adjusted series, and model-based seasonal adjustment components, saved in columns separated by white space

Table 5: **X-13ARIMA-SEATS File Extensions for Special SEATS Saved Output**

<i>Code</i>	<i>Description of table</i>
all	all series
xo	original series
n	seasonally adjusted series
s	seasonal factors
p	trend-cycle
u	irregular
c	transitory
cal	calendar
pa	preadjustment factor
cy	cycle
ltp	long term trend
er	residuals
rg0	separate regression component
rgsa	regression component in seasonally adjusted series
stp	stochastic trend cycle
stn	stochastic seasonally adjusted series
rtp	real time trend cycle
rtsa	real time seasonally adjusted series

Code gives the code used to specify the series in the `tabtables` argument of the `seats` spec.

Table 6: **Components Savable in .tbs file**

<i>Name</i>	<i>Short</i>	<i>Description of table</i>
componentmodels	mdc	models for the components
filtersaconc	fac	concurrent finite seasonal adjustment filter
filtersasym	faf	symmetric finite seasonal adjustment filter
filtertrendconc	ftc	concurrent finite trend filter
filtertrendsym	ftf	symmetric finite trend filter
pseudoinnovsadj	pia	pseudo-innovations of the final SEATS seasonal adjustment
pseudoinnovseasonal	pis	pseudo-innovations of the seasonal component
pseudoinnovtransitory	pit	pseudo-innovations of the transitory component
pseudoinnovtrend	pic	pseudo-innovations of the trend component
squaredgainsaconc	gac	squared gain for finite concurrent seasonal adjustment filter
squaredgainsasym	gaf	squared gain for finite symmetric seasonal adjustment filter
squaredgaintrendconc	gtc	squared gain for finite concurrent trend filter
squaredgaintrendsym	gtf	squared gain for finite symmetric trend filter
timeshiftsaconc	tac	time shift for finite concurrent seasonal adjustment filter
timeshifttrendconc	ttc	time shift for finite concurrent trend filter
wkendfilter	wkf	end filters of the semi-infinite Wiener-Kolmogorov filter
seasonaladjse	ase	standard error of final seasonally adjusted series
seasonalse	sse	standard error of final seasonal component
transitoryse	cse	standard error of final transitory component
trendse	tse	standard error of final trend component

Table 7: **Output Tables Available Only with save Argument for Seats**

<i>Name</i>	<i>Short</i>	<i>Description of table</i>
spectukeyorig	st0	Tukey spectral estimates of the first-differenced original series
spectukeysa	st1	Tukey spectral estimates of differenced, X-11 seasonally adjusted series (or of the logged seasonally adjusted series if <code>mode = logadd</code> or <code>mode = mult</code>)
spectukeyirr	st2	Tukey spectral estimates of outlier-modified X-11 irregular series
spectukeyseatssa	t1s	Tukey spectrum of the differenced final SEATS seasonal adjustment
spectukeyseatsirr	t2s	Tukey spectrum of the final SEATS irregular
spectukeyextresiduals	ter	Tukey spectrum of the extended residuals
spectukeyresidual	str	Tukey spectral estimates of the regARIMA model residuals
spectukeycomposite	it0	Tukey spectral estimates of first-differenced aggregate series
spectukeyindsa	it1	Tukey spectral estimates of the first-differenced indirect seasonally adjusted series
spectukeyindirr	it2	Tukey spectral estimates of outlier-modified irregular series from the indirect seasonal adjustment

Table 8: **Output Tables Available Only with save Argument for Spectrum**

Table 9: **Graphics Metafile Codes**

Code	Description
acf	residual autocorrelations
acf2	squared residual autocorrelations
adjcori	composite series (prior adjusted)
ador	original series (prior adjusted)
ahst	concurrent and revised seasonal adjustments and revisions
aichst	revision history of the likelihood statistics
ao	regARIMA AO outlier component
arat	final adjustment ratios
armahst	ARMA model coefficient history
bct	point backcasts and prediction intervals on the original scale
btr	point backcasts and standard errors for the transformed data
cad	regARIMA calendar adjusted original data
caf	combined adjustment factors

Table 9: **Graphics Metafile Codes (continued)**

Code	Description
cal	combined calendar adjustment factors
ccal	final combined calendar factors from irregular component regression
cfchst	forecast and forecast error history
chol	combined holiday component
chss	sliding spans of the changes in the seasonally adjusted series
cmpcad	regARIMA calendar adjusted composite data
cmpoad	regARIMA outlier adjusted composite data
cmpori	composite time series data (for the span analyzed)
cmppadj	prior adjusted composite data
cmppor	spectrum of the composite series
cmppsptukor	Tukey spectrum of the composite series
csahst	history of the period-to-period changes changes of the adjustments
ctd	final combined trading day factors from irregular component regression
ctrhst	history of the period-to-period changes of the trend-cycle values
fet	point forecasts and prediction intervals on the original scale
fethst	revision history of the out-of-sample forecasts
fintst	final outlier test statistics
fltsac	concurrent seasonal adjustment filter
fltsaf	symmetric seasonal adjustment filter
fltrnc	concurrent trend filter
fltrnf	symmetric trend filter
frfc	factors applied to get adjusted series with forced yearly totals
fttr	point forecasts and standard errors for the transformed data
idacf	residual autocorrelations for different orders of differencing
idpacf	residual partial autocorrelations for different orders of differencing
indahst	concurrent and revised indirect seasonal adjustments and revisions
indao	indirect additive outlier adjustment factors
indarat	indirect final adjustment ratios
indcaf	indirect combined adjustment factors
indcal	indirect calendar component
indchss	sliding spans of the changes in the indirect seasonally adjusted series
indfrfc	factors applied to get indirect adjusted series with forced yearly totals
indirr	indirect irregular component
indls	indirect level shift adjustment factors
indmirr	irregular component modified for extremes from indirect adjustment
indmori	original data modified for extremes from indirect adjustment
indmsa	seasonally adjusted data modified for extremes from indirect adjustment
indrsl	final replacement values for SI component of indirect adjustment
indsa	indirect seasonally adjusted data
indsar	rounded indirect final seasonally adjusted series
indsass	sliding spans of the indirect seasonally adjusted series
indsat	final indirect seasonally adjusted series with forced yearly totals
indsf	indirect seasonal component
indsfss	sliding spans of the indirect seasonal factors
indsi	indirect unmodified SI component
indspir	spectrum of indirect modified irregular component
indspsa	spectrum of differenced indirect seasonally adjusted series
indsptukir	Tukey spectrum of indirect modified irregular component
indsptuksa	Tukey spectrum of differenced indirect seasonally adjusted series
indtadj	indirect total adjustment factors
indtrn	indirect trend cycle
indyys	sliding spans of the year-to-year changes in the indirect seasonally adjusted series
irr	final irregular component
irrwt	final weights for irregular component
ls	regARIMA level shift outlier component
mdlest	regression and ARMA parameter estimates

Table 9: **Graphics Metafile Codes (continued)**

Code	Description
mirr	modified irregular series
mori	original data modified for extremes
msa	modified seasonally adjusted series
mvadj	original series adjusted for missing value regressors
oad	regARIMA outlier adjusted original data
ori	time series data (for the span analyzed)
oricnt	time series data plus constant (for the span analyzed)
orifctd	series forecast decomposition (SEATS)
otl	regARIMA combined outlier component
pacf	residual partial autocorrelation
padj	prior-adjusted data
padjt	prior-adjusted data (including prior trading day adjustments)
ppradj	permanent prior-adjusted data
ppradjt	permanent prior-adjusted data (including prior trading day adjustments)
pprior	permanent prior-adjustment factors
prior	prior-adjustment factors
ptd	prior trading day factors
regrsd	residuals from the estimated regression effects
rgseas	regARIMA user-defined seasonal component
rholiday	regARIMA holiday component
rsi	final replacement values for SI ratios
rtd	regARIMA trading day component
sa	final seasonally adjusted data
sac	final seasonally adjusted series with constant value added
safctd	final seasonally adjusted series forecast decomposition (SEATS)
sar	rounded final seasonally adjusted series
sass	sliding spans of the seasonally adjusted series
sat	final seasonally adjusted series with forced yearly totals
seataf	final combined adjustment factors (SEATS)
seatase	standard error of final seasonally adjusted series (SEATS)
seatcse	standard error of final transitory component (SEATS)
seatcyc	final cycle
seatdori	differenced original series after transformation, prior adjustment (SEATS)
seatsa	differenced final seasonally adjusted series (SEATS)
seatsdtr	differenced final trend (SEATS)
seatirr	final irregular component (SEATS)
seatirrotl	final irregular component outlier adjusted (SEATS)
seatltt	final long term trend
seatsa	final seasonally adjusted series (SEATS)
seatsaotl	final seasonally adjusted series adjusted for outliers (SEATS)
seatsf	final seasonal component (SEATS)
seatsse	standard error of final seasonal component (SEATS)
seatsm	sum of final seasonal component (SEATS)
seattrn	final trend component (SEATS)
seattse	standard error of final trend component (SEATS)
setarat	final adjustment ratios (SEATS)
setsac	final seasonally adjusted series with constant value added (SEATS)
settadj	total adjustment factors (SEATS)
settrc	final trend cycle with constant value added (SEATS)
settrns	final transitory component (SEATS)
sf	final seasonal factors
sffctd	final seasonal component forecast decomposition (SEATS)
sfhst	concurrent and projected seasonal component and their percent revisions
sfr	seasonal factors, adjusted for user-defined seasonal regARIMA component
sfss	sliding spans of the seasonal factors
sgsac	squared gain of the concurrent seasonal adjustment filter

Table 9: **Graphics Metafile Codes (continued)**

Code	Description
sgsaf	squared gain of the symmetric seasonal adjustment filter
sgtrnc	squared gain of the concurrent trend filter
sgtrnf	squared gain of the symmetric trend filter
si	final unmodified SI ratios
siox	final unmodified SI ratios, with labels for outliers and extreme values
so	regARIMA seasonal outlier component
spcsir	spectrum of the irregular component (SEATS)
spcssa	spectrum of the seasonally adjusted series (SEATS)
spctuksir	Tukey spectrum of the irregular component (SEATS)
spctukssa	Tukey spectrum of the seasonally adjusted series (SEATS)
spexrsd	spectrum of the extended residuals (SEATS)
spir	spectrum of modified irregular series
spor	spectrum of the original series
sprsd	spectrum of the regARIMA model residuals
spsa	spectrum of differenced seasonally adjusted series
sptukexrsd	Tukey spectrum of the extended residuals (SEATS)
sptukir	Tukey spectrum of modified irregular series
sptukor	Tukey spectrum of the original series
sptukrsd	Tukey spectrum of the regARIMA model residuals
sptuksa	Tukey spectrum of differenced seasonally adjusted series
tadj	total adjustment factors
tc	regARIMA temporary change outlier component
tdhst	trading day coefficient history
tdss	sliding spans of the trading day factors
tprior	temporary prior-adjustment factors
trancmp	regARIMA transitory component
tranfed	final transitory component forecast decomposition (SEATS)
trn	final trend cycle
trnfctd	final trend component forecast decomposition (SEATS)
trnhst	concurrent and revised Henderson trend-cycle values and revisions
tssac	time shift of the concurrent seasonal adjustment filter
tstrnc	time shift of the concurrent trend filter
usrdef	regARIMA user-defined regression component
xcal	final calendar factors from irregular component regression
xhol	final holiday factors from irregular component regression
xtd	final trading day factors from irregular component regression
xtrm	final extreme value adjustment factors
yyss	sliding spans of the year-to-year changes in the seasonally adjusted series